Science practice booklet: Grade 11 expectations

Grade 11 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 11 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Living things

Standards practised: HS-LS1-5, HS-LS1-6, HS-LS1-7, HS-LS2-3, HS-LS2-4, HS-LS2-5, HS-LS2-6, HS-LS2-7

A worked example

In a mangrove food chain, the producers capture 40000 kJ of energy from sunlight. The chain is: mangrove leaves and algae, then snails, then crabs, then herons. About 10 percent of the energy is passed on at each step. How much energy reaches the herons, and what happens to the rest?

1. There are three transfers: producers to snails, snails to crabs, crabs to herons.
2. Producers to snails: 10 percent of 40000 = 4000 kJ.
3. Snails to crabs: 10 percent of 4000 = 400 kJ.
4. Crabs to herons: 10 percent of 400 = 40 kJ.
5. The rest is used by each organism in cellular respiration to move, grow and stay alive, and most of it ends up as thermal energy released to the surroundings.

40 kJ reaches the herons; the rest is used in respiration or lost as heat.

Questions

1. Write the balanced chemical equation for photosynthesis. In what form is the energy stored afterwards?

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2. Aerobic respiration is C6H12O6 + 6O2 making 6CO2 + 6H2O + energy. Count the oxygen atoms on each side of the equation. What does your answer show about matter?

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3. In an ecosystem the producers capture 60000 kJ of energy. If about 10 percent is passed on at each step, how much energy is available to the secondary consumers (two steps up)?

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4. A population of secondary consumers needs 200 kJ of energy in the form of food. About 10 percent is passed on at each step. How much energy must the producers capture to support them? Why does this limit the number of top predators?

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5. Explain why cellular respiration releases energy, using the ideas of bonds breaking and bonds forming.

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6. A mangrove tree makes glucose by photosynthesis, but it also needs proteins. Explain how the tree can make proteins from glucose.

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7. Describe two processes that move carbon from living things or fuels into the atmosphere, and one process that removes it from the atmosphere.

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8. Mangrove forests store large amounts of carbon in their wood, roots and the mud around them. Explain why clearing a mangrove forest can add carbon dioxide to the atmosphere.

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9. A survey of a coastal area found 80 species of living things. Ten years later, after new building along the coast, a survey found 60 species. What percentage of the species was lost? Name one human activity that can reduce biodiversity.

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10. The Arabic oryx was once gone from the wild but was bred in protected reserves and then returned to its habitat. Give two reasons why protected areas and breeding programmes help to conserve a species.

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11. In a lake, a fish that eats small plant-eating animals is removed by overfishing. Predict what happens to the small plant-eating animals and then to the algae, and explain why.

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12. A town plans to build on a coast with a mangrove forest. Suggest a plan that reduces the impact on biodiversity, and name one criterion you would use to judge whether the plan works.

Section 2: Earth and space

Standards practised: HS-ESS2-4, HS-ESS2-6, HS-ESS3-1, HS-ESS3-2, HS-ESS3-3, HS-ESS3-4, HS-ESS3-5

A worked example

A reverse osmosis desalination plant produces 120000000 litres of fresh water a day. Suppose each person in a city uses 400 litres a day. How many people can the plant supply? Why does the process need a lot of energy?

1. Number of people = total water ÷ water per person.
2. 120000000 ÷ 400 = 300000 people.
3. In reverse osmosis, high-pressure pumps push seawater through a membrane that lets water through but holds back the salt.
4. Pumping at high pressure needs a lot of electrical energy.

300000 people; high pressure is needed to push the water through the membrane.

Questions

1. Sort these into renewable and non-renewable energy resources: solar, natural gas, wind, oil. Give one reason for each group.

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2. A solar park has a power output of 100 MW (100000 kW) for the equivalent of 6 hours of full sunshine a day. A home uses 30 kWh a day. About how many homes does the park supply in a day?

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3. Give one advantage and one disadvantage of generating electricity from natural gas, and one advantage and one disadvantage of generating it from solar panels.

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4. Explain how reverse osmosis makes fresh water from sea water. Name one possible effect of returning the very salty waste water (brine) to the sea.

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5. Sea water in a plant contains 40 g of salt in every litre. The plant takes in 1000 litres and produces 400 litres of fresh water, and all the salt stays in the rest. How many litres of brine are left, and what is its salt concentration in g per litre (to the nearest whole number)?

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6. Burning fossil fuels moves carbon between Earth's systems. Describe where the carbon was before and where it goes after, and say what happens when the ocean absorbs some of the extra carbon dioxide.

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7. Explain how greenhouse gases such as carbon dioxide and methane warm Earth's surface. Why is some greenhouse effect necessary for life?

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8. The amount of carbon dioxide in the air was about 280 parts per million before industry grew, and is now about 420 parts per million. What is the percentage increase? Name one kind of evidence that scientists use to know the past amounts.

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9. As Arctic ice melts, dark ocean water is uncovered. Explain why this can make warming faster. What do scientists call this kind of effect?

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10. Why do scientists use computer models to forecast climate, and why do they run them with several different scenarios of future emissions? Give one result that low-lying coasts need to plan for.

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11. An underground aquifer holds 800 million cubic metres of fresh water. People pump out 50 million cubic metres a year and only 10 million cubic metres a year flows back in. If this continues, in how many years will the aquifer be empty? Suggest one way to manage the water more sustainably.

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12. Suggest two ways a family can reduce its carbon dioxide emissions at home or when travelling. For each, explain how it reduces the amount of fossil fuel burned.

Section 3: Matter, forces and energy

Standards practised: HS-PS3-1, HS-PS3-2, HS-PS3-3, HS-PS3-4, HS-PS4-1, HS-PS4-3, HS-PS4-4, HS-PS4-5

A worked example

A student of mass 50 kg climbs a staircase that rises 4 m in 8 seconds. Use g = 10 N per kg. How much work does the student do against gravity, and what is the power?

1. Work = force × distance. The force needed to lift is the weight: 50 × 10 = 500 N.
2. Work = 500 N × 4 m = 2000 J.
3. Power = work ÷ time = 2000 J ÷ 8 s = 250 W.

2000 J of work; power 250 W.

Questions

1. A car of mass 1000 kg travels at 20 m per second. Use kinetic energy = ½ × mass × speed² to find its kinetic energy.

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2. A 2 kg ball is held 20 m above the ground (use g = 10 N per kg). Find its gravitational potential energy. If it is dropped and air resistance is ignored, find its speed just before it lands.

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3. A motor takes in 500 J of electrical energy and gives out 400 J of useful movement energy. What is its efficiency, and where does the other energy go?

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4. How much thermal energy is needed to heat 2 kg of water from 20 °C to 70 °C? Use Q = mass × specific heat capacity × temperature change, with a specific heat capacity of 4200 J per kg per °C.

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5. Name the main way heat is transferred in each case: (a) a metal spoon in a hot cup of karak gets hot, (b) warm air rises above a hot road, (c) you feel the warmth of the Sun on your face. In which direction does heat flow between two objects at different temperatures?

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6. A pump lifts 1000 kg of water through 10 m in 100 seconds (use g = 10 N per kg). Find the work done and the power of the pump.

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7. A radio station broadcasts at a frequency of 100 MHz (100000000 Hz). Radio waves travel at 300000000 m per second. Use speed = frequency × wavelength to find the wavelength.

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8. Suppose the Hope Probe is 180000000 km from Earth. How long does a radio signal from the probe take to reach Earth? Light and radio waves travel at 300000 km per second. Give the answer in seconds and in minutes.

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9. A solar panel of area 1.6 m^2 receives 1000 W of sunlight on every square metre. Its efficiency is 20 percent. What is its electrical power output?

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10. Explain in terms of photons how a solar panel produces an electric current.

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11. The energy of a photon rises with the frequency of the radiation. Use this idea to explain why ultraviolet light can damage skin cells but radio waves cannot.

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12. A digital signal sends information as a code of 0s and 1s. How many different values can 3 bits represent, and how many can 8 bits represent? Give one reason why digital signals are used in phones and optical fibre cables.

Answer guide for parents

Section 1: Living things

1. $6CO_2 + 6H_2O$, with energy from light, makes $C_6H_{12}O_6 + 6O_2$. The light energy is stored as chemical energy in the bonds of glucose.
2. Left side: 6 (in glucose) + 12 (in $6O_2$) = 18 oxygen atoms. Right side: 12 (in $6CO_2$) + 6 (in $6H_2O$) = 18. The numbers are equal because atoms are rearranged, not created or destroyed (conservation of matter).
3. 60000 × 0.1 = 6000 kJ for the primary consumers, then 6000 × 0.1 = 600 kJ for the secondary consumers.
4. 200 ÷ 0.1 = 2000 kJ for the primary consumers, then 2000 ÷ 0.1 = 20000 kJ for the producers. Each step up needs about ten times more energy below it, so only a small number of top predators can be supported.
5. Energy is needed to break the bonds in glucose and oxygen, and energy is released when the new bonds form in carbon dioxide and water. The new bonds hold less energy than the old ones, so there is a net release of energy that cells use to make ATP.
6. Glucose supplies carbon, hydrogen and oxygen. The tree takes in nitrogen (for example as nitrate dissolved in the water in the soil) and combines it with the atoms from glucose to build amino acids, which are joined into proteins.
7. Into the atmosphere: respiration (by plants, animals and microbes), decomposition, and burning of wood or fossil fuels. Out of the atmosphere: photosynthesis (or dissolving in the ocean).
8. When the trees are removed, the wood and the carbon-rich mud can decompose or be burned. Respiration and decay by microbes then release the stored carbon as carbon dioxide, and fewer trees remain to take carbon dioxide in by photosynthesis.
9. Lost: 80 − 60 = 20 species. 20 ÷ 80 = 0.25, so 25 percent. Activities include habitat loss from building, pollution, overfishing and introduced species.
10. Protected areas keep the habitat, food and water the species needs and limit hunting and disturbance. Breeding programmes increase the numbers and keep a wider variety of genes, so animals can be released into the wild to rebuild the population.
11. The small plant-eating animals increase because fewer are eaten. They then eat more algae, so the amount of algae falls. Removing one species changes the populations of the others because they are linked in the food web.
12. A possible plan: keep a protected strip of mangroves, build away from nursery areas, and plant new mangrove seedlings elsewhere. Criteria: the number of species counted before and after, the area of mangrove kept, or the water quality measured each year.

Section 2: Earth and space

1. Renewable: solar and wind, because they are replaced continually by the Sun and the atmosphere. Non-renewable: natural gas and oil, because they formed over millions of years and are used up far faster than they form.
2. Energy = 100000 kW × 6 h = 600000 kWh. 600000 ÷ 30 = 20000 homes.

3. Natural gas: reliable at any time and can be switched on when needed; but it releases carbon dioxide and the supply is limited. Solar: no carbon dioxide while generating and the fuel is free; but output stops at night and varies with clouds and dust, so storage or back-up is needed and large areas of land are used.
4. High pressure pushes sea water through a membrane that lets water pass but holds back most of the dissolved salt. The brine is saltier and can harm sea life near the outlet if it is not mixed and spread out well.
5. Brine = 1000 − 400 = 600 litres. All the salt: 1000 × 40 = 40000 g. 40000 ÷ 600 = 66.7, so about 67 g per litre.
6. The carbon was stored in the geosphere (underground as coal, oil and gas) and goes into the atmosphere as carbon dioxide. When the ocean absorbs some of it, the sea water becomes more acidic, which makes it harder for corals and shellfish to build their shells and skeletons.
7. Earth's surface gives out infrared radiation. Greenhouse gases absorb some of it and send it out in all directions, including back down, so the lower atmosphere and surface stay warmer. Without any greenhouse effect, Earth would be far too cold for liquid water and most life.
8. Increase = 420 − 280 = 140. 140 ÷ 280 = 0.5, so a 50 percent increase. Evidence: air bubbles trapped in ice cores (also tree rings, sediments and coral records).
9. Ice reflects most sunlight, but dark water absorbs more of it. More absorbed energy warms the water, which melts more ice, which uncovers more dark water. This is a positive feedback (a change that makes itself bigger).
10. A model puts together many measured data and the physical laws for the air, ocean, ice and living things, so scientists can test what may happen over decades, which cannot be done in a laboratory. Different scenarios show how the results depend on human choices. Rising sea level (from melting land ice and the expansion of warmer water) is a result that low-lying coasts must plan for.
11. Net loss = 50 − 10 = 40 million cubic metres a year. 800 ÷ 40 = 20 years. Ways: use less water (efficient irrigation), reuse treated waste water, or use desalinated water to reduce pumping.
12. Examples: raise the air-conditioner setting a little and use efficient appliances, so the home needs less electricity from power stations that burn fuel; share car journeys or use the metro or a bus, so fewer litres of fuel are burned for each person.

Section 3: Matter, forces and energy

1. ½ × 1000 × 20^2 = ½ × 1000 × 400 = 200000 J
2. Potential energy = mgh = 2 × 10 × 20 = 400 J. All of it becomes kinetic energy: ½ × 2 × v^2 = 400, so v^2 = 400 and v = 20 m per second.
3. Efficiency = 400 ÷ 500 = 0.8, or 80 percent. The other 100 J is wasted, mostly as thermal energy (heat) and some sound, and it spreads out into the surroundings. Total energy is still conserved.
4. Q = 2 × 4200 × (70 − 20) = 2 × 4200 × 50 = 420000 J
5. (a) Conduction. (b) Convection. (c) Radiation. Heat always flows from the hotter object to the cooler one until they reach the same temperature (thermal equilibrium).

6. Weight = 1000 × 10 = 10000 N. Work = 10000 × 10 = 100000 J. Power = 100000 ÷ 100 = 1000 W.
7. Wavelength = speed ÷ frequency = 300000000 ÷ 100000000 = 3 m
8. 180000000 ÷ 300000 = 600 seconds, which is 10 minutes. Radio waves are electromagnetic waves, so they can cross empty space.
9. Input power = 1.6 × 1000 = 1600 W. Output = 20 percent of 1600 = 0.2 × 1600 = 320 W.
10. Light arrives as photons, packets of energy. When a photon is absorbed in the semiconductor (such as silicon), it gives its energy to an electron and frees it. The panel's structure pushes these freed electrons in one direction, which makes an electric current.
11. Ultraviolet photons have a much higher frequency than radio photons, so each photon carries enough energy to break chemical bonds in molecules in cells. A radio photon carries far too little energy, however many of them there are.
12. 3 bits: $2^3 = 8$ values. 8 bits: $2^8 = 256$ values. Digital signals are less affected by noise because the receiver only has to decide between 0 and 1, errors can be detected and corrected, and the signal can be copied and sent over long distances without loss of quality.